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Research Title: Understanding Groundwater-Dependent Drought Adaptation And Agricultural Resilience: A Mixed-Methods Study Among Smallholder Farmers In Northwestern Bangladesh

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UNDERSTANDING GROUNDWATER-DEPENDENT DROUGHT ADAPTATION AND AGRICULTURAL RESILIENCE: A MIXED-METHODS STUDY AMONG SMALLHOLDER FARMERS IN NORTHWESTERN BANGLADESH

Introduction:

Smallholder agriculture is increasingly challenged by climate change, as rising temperatures, erratic rainfall, and recurring droughts place growing pressure on water-dependent farming systems [3], [5]. In this context, agricultural resilience involves more than recovering from a single climatic shock; it also depends on farmers' ability to retain the resources, flexibility, and capacity needed to cope with future challenges. Drought poses a particular challenge because prolonged rainfall shortages and reduced water availability can progressively disrupt crop production. For smallholder farmers, irrigation is therefore an important adaptation strategy, reducing dependence on uncertain rainfall and helping sustain crop production during dry periods [2], [3].

However, when irrigation relies mainly on groundwater, the sustainability of this adaptation becomes closely tied to the condition of the resource. This issue is especially important in northwestern Bangladesh, where recurrent drought, limited surface water, and intensive dry-season cultivation have made groundwater an important source of irrigation [1], [2]. Previous research has reported groundwater depletion and declining water availability in the drought-prone Barind region [1], [4]. Recent evidence also identifies groundwater depletion as a major drought-related impact, while groundwater extraction remains a common coping response among farmers [5]. Groundwater irrigation therefore presents a potential trade-off: it can protect crops and farm income in the short term, but increasing dependence may also raise irrigation costs and expose farmers to greater water-resource constraints in the future [2], [5]. Understanding this relationship is important for assessing the longer-term sustainability of drought adaptation.

Although Bangladesh has a substantial body of research on climate hazards, farmer adaptation, and groundwater resources, these dimensions have often been examined separately [2], [3], [5]. A systematic review of 106 adaptation studies found that only 5% explicitly considered adaptation through a social–ecological systems perspective, with limited attention to adaptation outcomes and interactions between household responses and the resources supporting them [6]. Similarly, existing studies in the Barind region have documented drought impacts, groundwater stress, and farmers' coping strategies [1],[3],[5], but relatively less attention has been given to how dependence on groundwater as an adaptation pathway relates to farmers' longer-term agricultural resilience. This study addresses this gap by examining the relationship between groundwater-dependent drought adaptation and agricultural resilience among smallholder farmers in northwestern Bangladesh. By connecting farmers' adaptation decisions with the groundwater resources on which those decisions depend, the study aims to generate evidence for more sustainable irrigation management, improved groundwater governance, and climate-resilient agricultural planning in drought-prone areas of Bangladesh.

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Research Questions:

1. To what extent do smallholder farmers depend on groundwater for drought adaptation in northwestern Bangladesh?
2. How is groundwater dependence associated with agricultural resilience among smallholder farmers?
3. What factors explain the relationship between groundwater dependence and agricultural resilience?
4. How does groundwater dependence relate to irrigation costs, production stability, and farmers' adaptive flexibility?
5. How do farmers perceive the long-term sustainability of groundwater-dependent drought adaptation?

Literature Review: Drought is a recurring challenge for smallholder agriculture in northwestern Bangladesh, where limited surface-water availability has made groundwater irrigation central to dry-season crop production [1], [2]. Earlier studies have documented farmers' drought-adaptation practices, water scarcity, and groundwater depletion across the region [3], [4]. More recent evidence from the Barind Tract (northwestern Bangladesh) further identifies groundwater depletion as a major drought impact and groundwater extraction as a common coping response [5]. Together, these studies establish groundwater as an important component of farmers' drought adaptation, while also highlighting increasing pressure on the resource itself. However, existing research has largely examined adaptation practices, agricultural outcomes, and groundwater conditions as separate issues. Studies on farmer adaptation primarily focus on the strategies households adopt in response to drought, whereas groundwater research emphasizes declining water levels, extraction, and resource sustainability [1], [3], [5]. Consequently, less is known about whether dependence on groundwater is associated with farmers' ability to maintain production, absorb future shocks, and adjust their farming decisions. This distinction is important because groundwater may provide immediate protection against drought while increasing irrigation costs, resource dependence, and vulnerability to future water shortages. This limitation is also evident in the broader Bangladesh adaptation literature. A systematic review of 106 studies found that only 5% explicitly considered adaptation through a social–ecological systems perspective, with relatively limited attention to adaptation outcomes and interactions between household responses and the ecological resources supporting them [6]. The literature therefore provides substantial evidence on what farmers experience and how they respond, but less evidence on how an adaptation strategy may shape the resilience of the farming system over time.

These findings directly inform the proposed study. Rather than repeating a general assessment of drought-coping practices, the research will examine the extent of groundwater dependence, its relationship with agricultural resilience, and the factors that shape this relationship among smallholder farmers in northwestern Bangladesh. A mixed-methods design is appropriate because household-level quantitative data can identify patterns in groundwater dependence, production stability, irrigation costs, and adaptive flexibility, while qualitative evidence can explain farmers'

irrigation decisions, constraints, and perceptions of future water availability. By connecting farmers' adaptation choices with the resource base on which those choices depend, the study seeks to address an important gap in understanding groundwater-dependent adaptation and its implications for longer-term agricultural resilience.

Methodology: The study will adopt a realist, mixed-methods approach to examine the relationship between groundwater-dependent drought adaptation and agricultural resilience. A realist perspective is appropriate because the study considers both measurable conditions such as groundwater dependence, irrigation costs, and production stability, and the social and economic factors that shape farmers' decisions and experiences. The research will use an explanatory sequential design, beginning with a household survey and followed by qualitative interviews to explain the patterns emerging from the quantitative findings. The study will focus on smallholder farming households in selected drought-prone areas of northwestern Bangladesh that depend on irrigation for crop production. A multistage sampling procedure will be used: drought-prone localities will be purposively identified based on documented water stress, followed by village selection and random selection of eligible farming households. Approximately 250–300 households will be surveyed. From the survey participants, 20–25 farmers representing different levels of groundwater dependence will be purposively selected for semi-structured interviews. This combination will provide both broad household-level evidence and detailed accounts of farmers' experiences while keeping the study feasible within the available time and resources. Primary quantitative data will cover the extent of groundwater dependence, irrigation frequency and expenditure, crop production stability, crop-choice flexibility, farm characteristics, and relevant household and institutional factors. Agricultural resilience will be assessed through clearly defined dimensions including the ability to maintain production during drought, economic capacity to sustain farming, and flexibility in adjusting farm practices. Qualitative interviews will explore farmers' irrigation decisions, changes in water availability, economic constraints, experiences during drought, and perceptions of future water security. Survey data will be analysed using descriptive statistics, correlation, and multivariable regression to examine the relationship between groundwater dependence and agricultural resilience while accounting for relevant influencing factors. Interview data will be analysed thematically to identify recurring explanations, constraints, and decision-making patterns. Findings from both components will then be integrated to identify where quantitative relationships are supported, challenged, or explained by farmers' experiences. The questionnaire will be pre-tested before the main survey to improve clarity and reliability. Informed consent, voluntary participation, confidentiality, and secure data handling will be maintained throughout. Triangulation across quantitative and qualitative evidence will strengthen the credibility of the findings. As the study will be observational and cross-sectional, results will be interpreted primarily as associations rather than causal effects.

Project Timeline:

Phase	Activities	Duration	Milestones
Phase 1: Preparation	Literature review, conceptual framework, research questions, site selection and sampling design	Month 1	Finalized research framework and study plan
Phase 2: Tool Development	Questionnaire and interview-guide development, pre-testing and revision	Month 2	Finalized data-collection tools
Phase 3: Field Data Collection	Household survey and semi-structured interviews	Months 2–3	Completion of primary data collection
Phase 4: Data Processing & Analysis	Data cleaning, coding, statistical analysis, and thematic analysis	Months 3–5	Completed quantitative and qualitative analyses
Phase 5: Integration & Interpretation	Integration of findings and interpretation of groundwater dependence–resilience relationships	Month 5	Consolidated research findings
Phase 6: Reporting & Dissemination	Report writing, final review, and dissemination of findings	Month 6	Final research report and dissemination outputs

Limitations:

1. Cross-sectional design: Shows associations, not causality.
2. Recall bias: Past production, irrigation costs, and water use may be imperfectly recalled.
3. Self-reported data: Some information may rely on farmers' estimates and perceptions.
4. Limited records: Reliable household-level groundwater-use data may be unavailable.
5. Geographical scope: Findings may not represent all of northwestern Bangladesh.
6. Time and resources: Limited fieldwork time and budget may restrict seasonal coverage and sample size.

Conclusion: This study will examine how groundwater-dependent drought adaptation is associated with agricultural resilience among smallholder farmers in northwestern Bangladesh. By linking farmers' reliance on groundwater with production stability, irrigation costs, and adaptive capacity, the research will provide a broader understanding of the benefits and pressures associated with groundwater-based adaptation. The mixed-methods approach will combine measurable household-level patterns with farmers' experiences and perceptions, helping explain the factors shaping resilience under water stress. The findings are expected to provide practical evidence for more sustainable irrigation practices, informed water-resource management, and climate-resilient agricultural planning in drought-prone areas of Bangladesh.

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